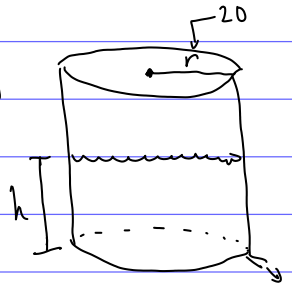
9




$r = 20$ (constant)

$h$

$V = \pi r^2 h$

$V = \pi (20)^2 h$

$V = 400\pi h$

$\frac{dV}{dt} = -25$

$\downarrow \frac{d}{dt}$

Find $\frac{dh}{dt}$

$\frac{dV}{dt} = 400\pi \cdot \frac{dh}{dt}$

$-25 = 400\pi \cdot \frac{dh}{dt}$

$\frac{-25}{400\pi} = \frac{dh}{dt}$

$\frac{-1}{16\pi} = \frac{dh}{dt}$

The height of the water decreases at the rate of $\frac{1}{16\pi}$ cm/sec.